Low-Cost Educational Counter-Unmanned Aerial System (C-UAS) Prototype

Proposed Block Diagram

```
None
[ 24 GHz mmWave Radar (LD2410) ]
              | (UART Telemetry)
              ▼
[ ESP32 Microcontroller (Central Unit) ]
         |                        |
  (PWM Control)           (Digital Trigger)
         ▼                        ▼
[ Pan-Tilt Servos (2-DOF) ]  [ Neutralization Unit (Jammer / EMP) ]
```

Methodology

The proposed system follows a modular **Detect–Track–Engage** architecture using commercial off-the-shelf components to demonstrate a real-world C-UAS workflow at a small scale:

- **Detection Stage:** A 24 GHz mmWave radar module (LD2410) continuously scans the local airspace, transmitting microwave signals and analyzing reflected echoes. It provides real-time distance, relative velocity (Doppler shift), and direction data over UART to an ESP32 microcontroller.
- **Target Locking & Tracking:** The ESP32 processes the telemetry to filter target trajectories based on motion and range. Upon target identification, a closed-loop control algorithm calculates azimuth and elevation vectors to drive a 2-DOF pan-tilt servo mechanism, maintaining continuous directional alignment with the target.
- **Neutralization Stage:** A directional effector mounted on the pan-tilt mechanism executes one of two lab-safe countermeasures:
    - **RF Jamming (Soft-Kill):** Transmits targeted low-power signals in the 2.4 GHz ISM band via an ESP32 or CC1101 module to interrupt the drone's command-and-control link, inducing failsafe behavior.

- **Electromagnetic Pulse (EMP Demo):** Triggers a controlled, low-energy pulse circuit (capacitor discharge bank and coil via MOSFET) to induce short-range interference for test-circuit disruption in a shielded setup.

Proposed System Architecture

Subsystem Module	Core Hardware	Data / Control Line	Primary Function
Detection Unit	24 GHz mmWave Radar (LD2410)	UART Telemetry	Scans local airspace; outputs distance, velocity (Doppler), and directional data
Central Processing	ESP32 Microcontroller	Processing & Control Hub	Filters target trajectories, computes 2-DOF alignment vectors, and issues trigger commands
Tracking Mechanism	2-DOF Pan-Tilt Servos	PWM Signals	Dynamically adjusts azimuth and elevation to maintain active orientation toward the target
Neutralization Unit	2.4 GHz RF Module / Capacitor Bank	Digital Trigger	Executes controlled RF jamming or localized low-energy pulse disruption

Proposed Outcome

- **Working Educational Prototype:** A functional, low-cost (₹3,000–6,000) small-scale C-UAS hardware stack validating multi-stage electronic defense concepts.
- **Closed-Loop System Integration:** Seamless fusion of mmWave radar detection, microcontroller decision-making, servo-driven target tracking, and electronic countermeasures.
- **Hands-On Academic Platform:** A practical testbed for studying radar signal processing, control loops, RF link disruption, and electronic warfare principles.
- **Extensible Framework:** A scalable baseline allowing future enhancements, including camera-based computer vision fusion, AI/ML drone classification, and multi-frequency jamming.